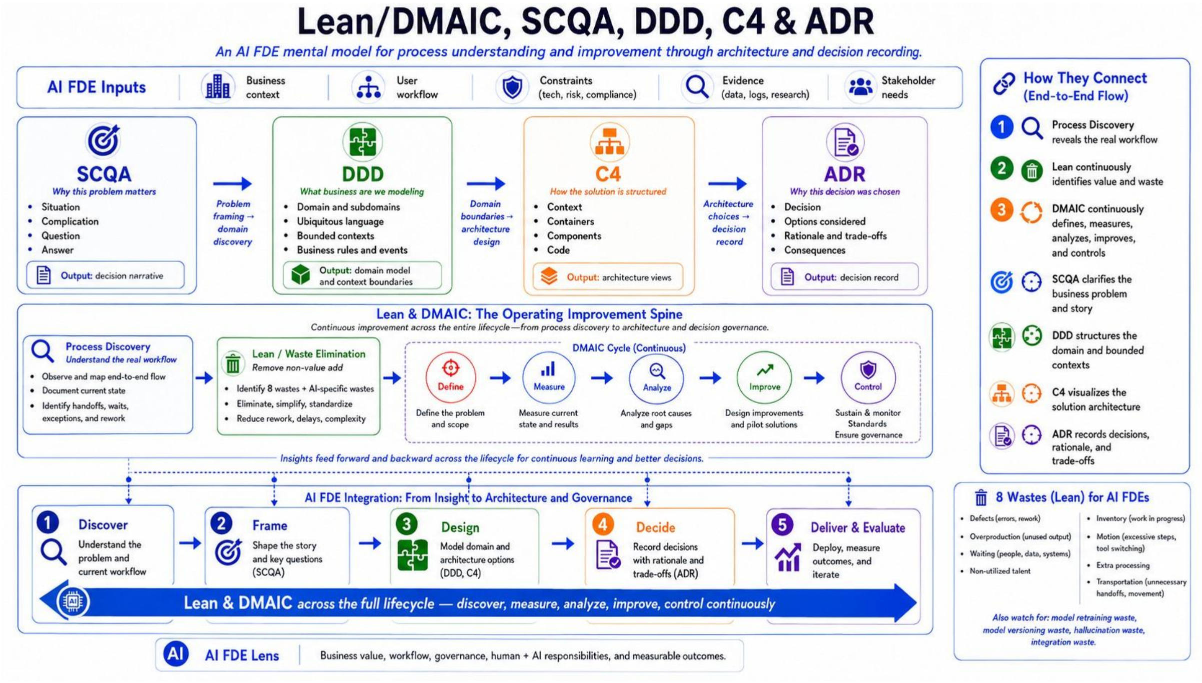
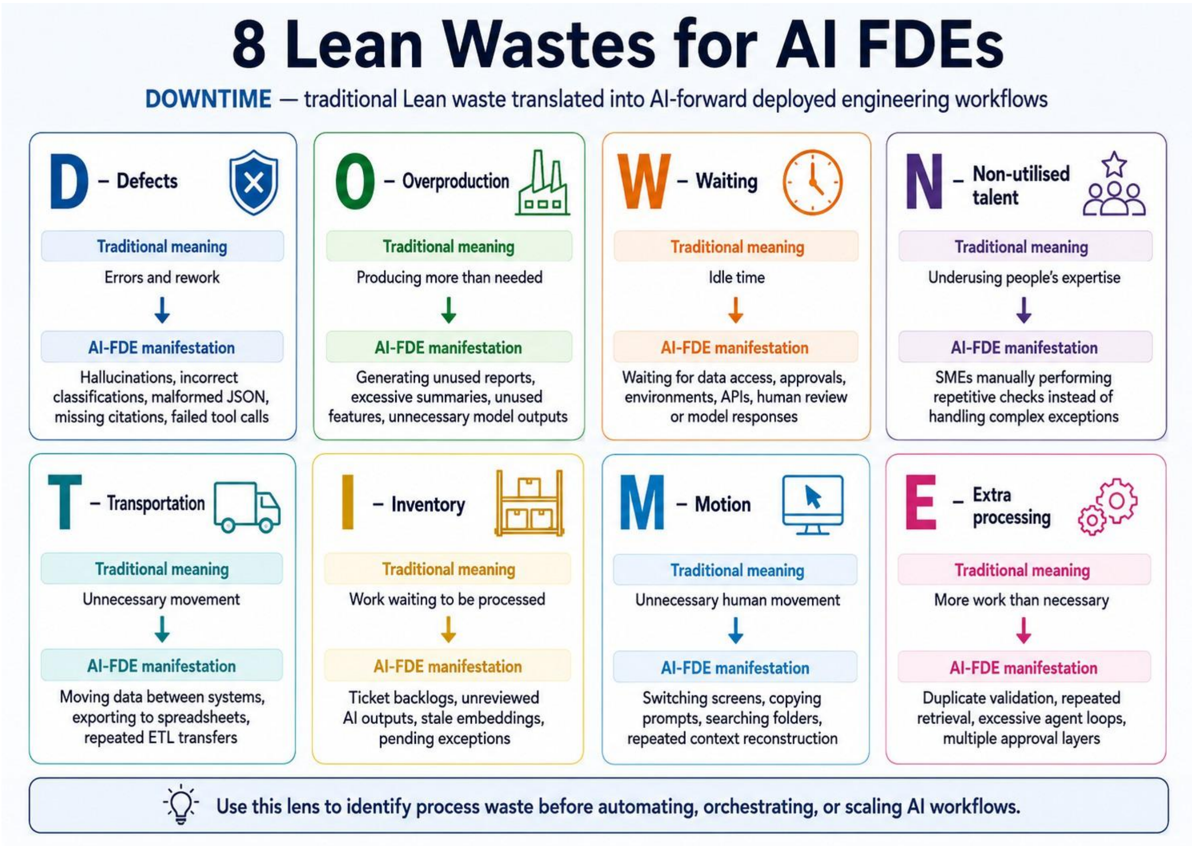
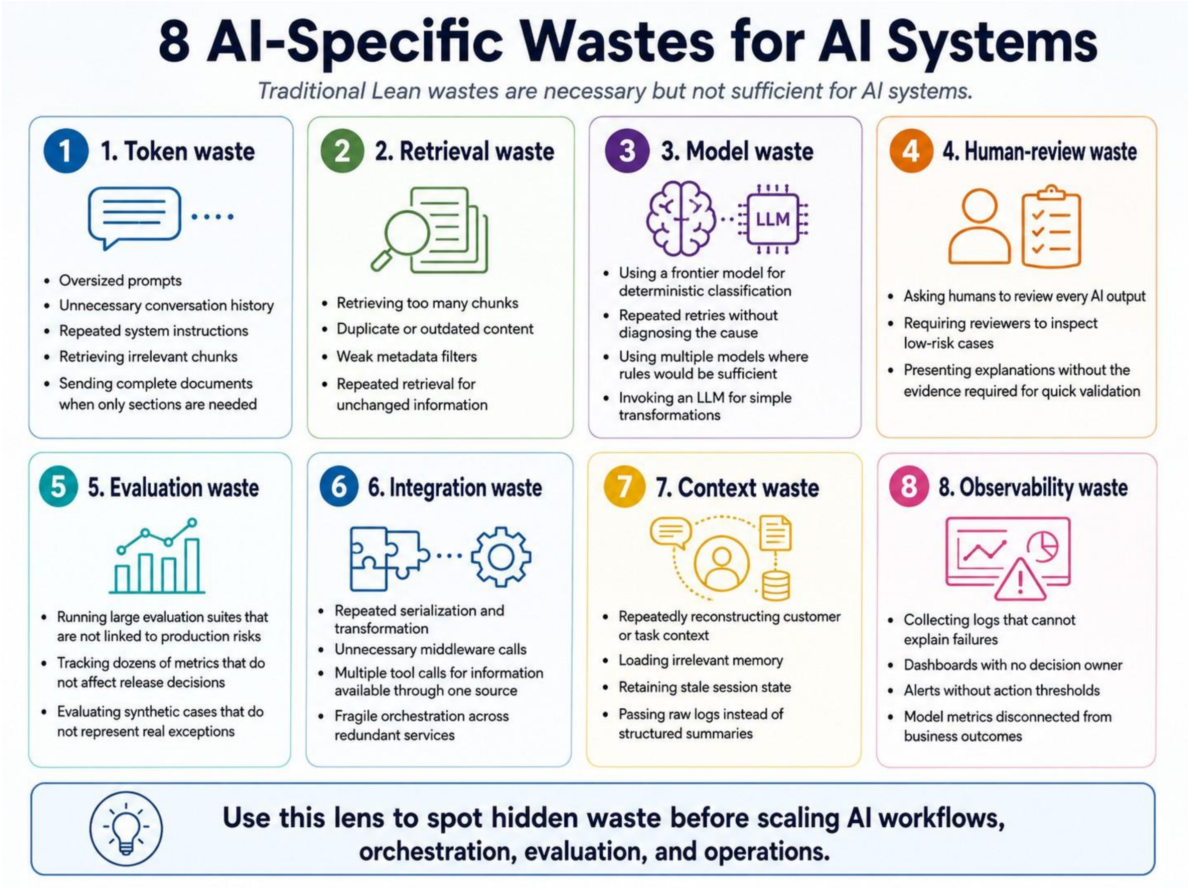
Lean - DMAIC


Lean/DMAIC, SCQA, DDD, C4 & ADR
An AI FDE mental model for process understanding and improvement through architecture and decision recording.
AI FDE Inputs
Business context
User workflow
Constraints (tech, risk, compliance)
Evidence (data, logs, research)
Stakeholder needs
SCQA
Why this problem matters
• Situation
• Complication
• Question
• Answer
Output: decision narrative
Problem framing → domain discovery
DDD
What business are we modeling
• Domain and subdomains
• Ubiquitous language
• Bounded contexts
• Business rules and events
Output: domain model and context boundaries
Domain boundaries → architecture design
C4
How the solution is structured
• Context
• Containers
• Components
• Code
Output: architecture views
Architecture choices → decision record
ADR
Why this decision was chosen
• Decision
• Options considered
• Rationale and trade-offs
• Consequences
Output: decision record
Lean & DMAIC: The Operating Improvement Spine
Continuous improvement across the entire lifecycle—from process discovery to architecture and decision governance.
Process Discovery
Understand the real workflow
• Observe and map end-to-end flow
• Document current state
• Identify handoffs, waits, exceptions, and rework
Lean / Waste Elimination
Remove non-value add
• Identify 8 wastes + AI-specific wastes
• Eliminate, simplify, standardize
• Reduce rework, delays, complexity
DMAIC Cycle (Continuous)
Define
Define the problem and scope
Measure
Measure current state and results
Analyze
Analyze root causes and gaps
Improve
Design improvements and pilot solutions
Control
Sustain & monitor Standards Ensure governance
Insights feed forward and backward across the lifecycle for continuous learning and better decisions.
AI FDE Integration: From Insight to Architecture and Governance
1 Discover
Understand the problem and current workflow
2 Frame
Shape the story and key questions (SCQA)
3 Design
Model domain and architecture options (DDD, C4)
4 Decide
Record decisions with rationale and trade-offs (ADR)
5 Deliver & Evaluate
Deploy, measure outcomes, and iterate
Lean & DMAIC across the full lifecycle — discover, measure, analyze, improve, control continuously
AI FDE Lens
Business value, workflow, governance, human + AI responsibilities, and measurable outcomes.
How They Connect (End-to-End Flow)
1 Process Discovery reveals the real workflow
2 Lean continuously identifies value and waste
3 DMAIC continuously defines, measures, analyzes, improves, and controls
SCQA clarifies the business problem and story
DDD structures the domain and bounded contexts
C4 visualizes the solution architecture
ADR records decisions, rationale, and trade-offs
8 Wastes (Lean) for AI FDEs
• Defects (errors, rework)
• Overproduction (unused output)
• Waiting (people, data, systems)
• Non-utilized talent
• Inventory (work in progress)
• Motion (excessive steps, tool switching)
• Extra processing
• Transportation (unnecessary handoffs, movement)
Also watch for: model retraining waste, model versioning waste, hallucination waste, integration waste.


8 Lean Wastes for AI FDEs
DOWNTIME — traditional Lean waste translated into AI-forward deployed engineering workflows
D – Defects
Traditional meaning
Errors and rework
AI-FDE manifestation
Hallucinations, incorrect classifications, malformed JSON, missing citations, failed tool calls
O – Overproduction
Traditional meaning
Producing more than needed
AI-FDE manifestation
Generating unused reports, excessive summaries, unused features, unnecessary model outputs
W – Waiting
Traditional meaning
Idle time
AI-FDE manifestation
Waiting for data access, approvals, environments, APIs, human review or model responses
N – Non-utilised talent
Traditional meaning
Underusing people's expertise
AI-FDE manifestation
SMEs manually performing repetitive checks instead of handling complex exceptions
T – Transportation
Traditional meaning
Unnecessary movement
AI-FDE manifestation
Moving data between systems, exporting to spreadsheets, repeated ETL transfers
I – Inventory
Traditional meaning
Work waiting to be processed
AI-FDE manifestation
Ticket backlogs, unreviewed AI outputs, stale embeddings, pending exceptions
M – Motion
Traditional meaning
Unnecessary human movement
AI-FDE manifestation
Switching screens, copying prompts, searching folders, repeated context reconstruction
E – Extra processing
Traditional meaning
More work than necessary
AI-FDE manifestation
Duplicate validation, repeated retrieval, excessive agent loops, multiple approval layers
Use this lens to identify process waste before automating, orchestrating, or scaling AI workflows.


8 AI-Specific Wastes for AI Systems
Traditional Lean wastes are necessary but not sufficient for AI systems.
1. Token waste
Oversized prompts
Unnecessary conversation history
Repeated system instructions
Retrieving irrelevant chunks
Sending complete documents when only sections are needed
2. Retrieval waste
Retrieving too many chunks
Duplicate or outdated content
Weak metadata filters
Repeated retrieval for unchanged information
3. Model waste
LLM
Using a frontier model for deterministic classification
Repeated retries without diagnosing the cause
Using multiple models where rules would be sufficient
Invoking an LLM for simple transformations
4. Human-review waste
Asking humans to review every AI output
Requiring reviewers to inspect low-risk cases
Presenting explanations without the evidence required for quick validation
5. Evaluation waste
Running large evaluation suites that are not linked to production risks
Tracking dozens of metrics that do not affect release decisions
Evaluating synthetic cases that do not represent real exceptions
6. Integration waste
Repeated serialization and transformation
Unnecessary middleware calls
Multiple tool calls for information available through one source
Fragile orchestration across redundant services
7. Context waste
Repeatedly reconstructing customer or task context
Loading irrelevant memory
Retaining stale session state
Passing raw logs instead of structured summaries
8. Observability waste
Collecting logs that cannot explain failures
Dashboards with no decision owner
Alerts without action thresholds
Model metrics disconnected from business outcomes
Use this lens to spot hidden waste before scaling AI workflows, orchestration, evaluation, and operations.